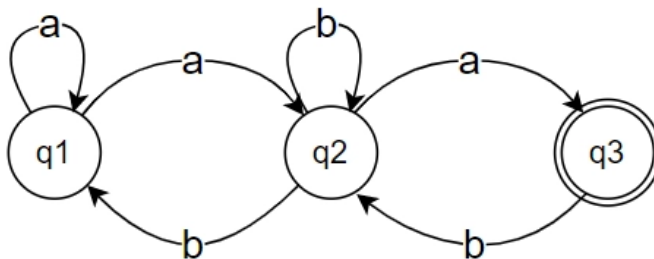
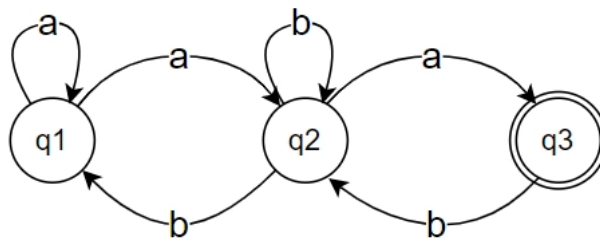
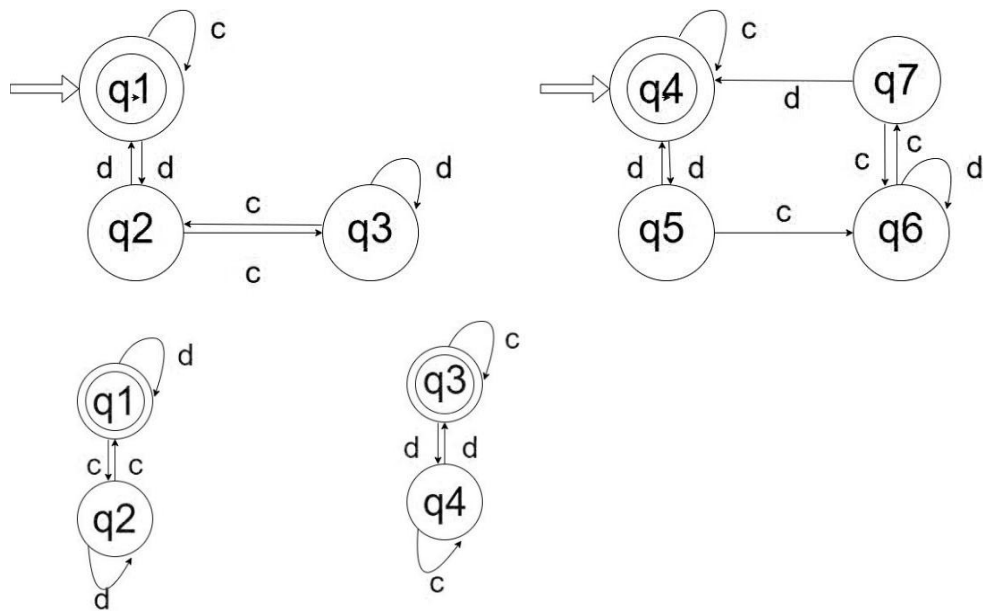


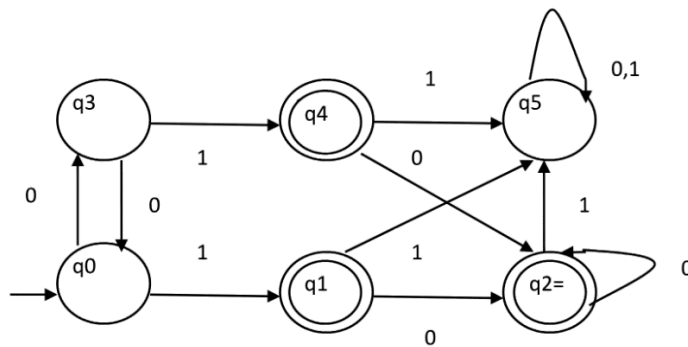
1. Define Turing Machine with formal definition and diagram.
2. What is Universal Turing Machine.
3. Describe variants of Turing Machines and their equivalence.
4. Show equivalence between CFG and PDA.
5. Explain DPDA vs NPDA.
6. Construct PDA for a given context-free language $L = \{ a^n b^n \mid n \geq 0 \}$
7. Check whether they are equivalent or not.
8. Design an NFA from the given RE $[a (a^* b a^* b a^*)^*]$.
9. Design Finite Automata from the given Regular Expressions
 $a+b$
 ab
 $[ab + (b + aa)b^* a]$
10. What are regular expressions?
11. Prove that $0^*1(0+10^*1)^*$ is equal to $(1+00^*1)+(1+00^*1)(0+10^*1)^*(0+10^*1)$
12. Convert the following NFA to a regular expression.



13. What is an equivalent DFA?
14. Check whether they are equivalent or not.



15. Minimize the following DFA



16. Explain parse trees and their relation to derivations.

17. Explain regular vs context-free grammars with differences.

18. Discuss ambiguity in grammars with examples.

19. Generate some strings by using this Language $L = \{ \text{Set of all strings over } \{0,1\} \text{ containing '00'} \}$

20. Construct a NFA by using the above Language then Convert the NFA to its equivalent DFA.

21. Explain the Chomsky Hierarchy with examples of each type of grammar.

22. Explain different operations on languages.

23. Differentiate between DFA and Turing Machine.

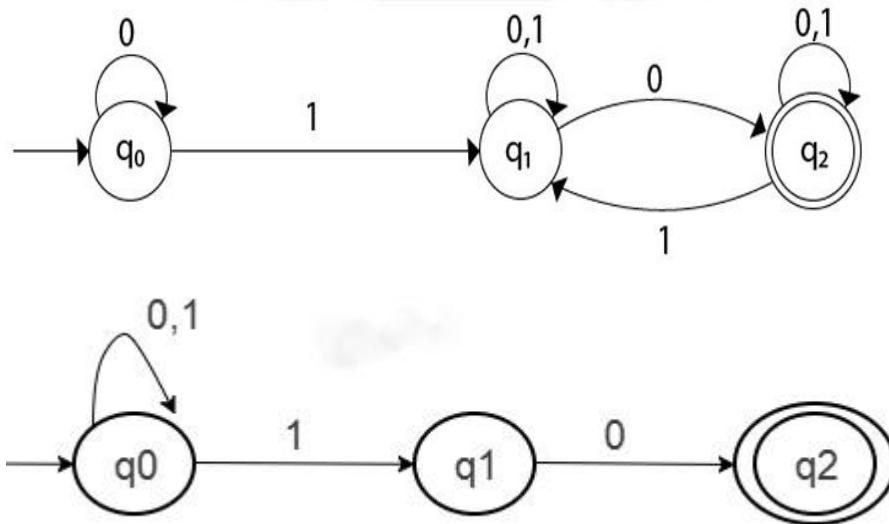
24. Define a Turing Machine (TM). Explain its components.

25. Construct a PDA for language $L = \{a^n b^n \mid n \geq 1\}$.

26. Define Pushdown Automaton (PDA). Explain its components.

27. Explain derivation in CFG. Differentiate between leftmost and rightmost derivation.

28. Convert the given NFA to its equivalent DFA.



29. Convert the regular expression $(a + b)^*ab$ into an equivalent DFA.

Prove that the class of regular languages is closed under intersection.

30. Construct a finite state machine that accepts the language, over $\{0,1\}$, of all strings that contain neither the substring 00 nor the substring 11.

31. Which of the following Finite Automaton is having ambiguity and why? i) NFA ii) DFA
Design a DFA that accepts set of all strings with abab as a substring over alphabet $\{a, b\}$.

32. Draw Transition diagram for the Transition table given below and check acceptance for the strings: i) 1001 ii) 0001

33. Define Finite Automaton(FA). Explain in detail different ways to represent it.
Differentiate between NFA and DFA.

34. Define

(i) String

(ii) Alphabet

(iii) Language

(iv) Closure

(v) Powerset

35. Given a grammar, construct both leftmost and rightmost derivations for a given string and draw the parse tree.

36. Explain types of grammars in Chomsky hierarchy.

Prove that Kleene star always contains the empty string (ϵ) .

37. Explain operations on strings: concatenation, length, and reverse with examples.

Let $L1=\{a,ab\}$, $L2=\{b,ba\}$ Find:

$L1 \cup L2$

$L1L2$

Apply Rice's Theorem to prove that "Language accepted by TM is empty" which is

undecidable.

38. Explain the Halting Problem and prove that it is undecidable.

Define decidable and undecidable problems. Explain with examples.

39. Compare Turing Machine with DFA and PDA. Why is TM more powerful?

40. Define Turing Machine with its components.

Prove that the language $L = \{ww^R \mid w \in \{a, b\}^*\}$ cannot be accepted by any DPDA but can be accepted by an NPDA.

41. Design a PDA for the language:

$$L = \{a^n b^n \mid n \geq 0\}.$$

42. Eliminate left recursion from the grammar:

$$S \rightarrow S a \mid b$$

and convert it into an equivalent grammar.

43. Prove that the grammar is ambiguous or unambiguous:

$$E \rightarrow E + E \mid E * E \mid id$$

44. What is a parse tree?

Construct a parse tree for the string "aaba" using a suitable CFG.

45. Define Context-Free Grammar (CFG). Explain its components with an example.

46. Explain the difference between left linear grammar and right linear grammar with examples.

47. Prove that $L = \{a^i 2 \mid i \geq 1\}$ is not regular.

Find regular expression for the following DFA using Arden's Theorem-

48. Define Regular Language and Regular Grammar. Give one example of each.

49. What are closure properties of regular languages? Explain closure under:

a) Union

b) Concatenation

c) Kleene Star

50. Construct a regular grammar for the language:

$$L = \{a^n b \mid n \geq 1\}$$

51. Convert a given Moore machine into equivalent Mealy machine and justify.

52. Is every NFA an ϵ -NFA? Justify. Draw the transition diagram of an NFA over $\Sigma = \{0, 1\}$ that accepts all strings of length at least 2.

53. Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA). Show transition table and transition graph.

54. Write the notations of DFA. Give the DFA accepting the language over the alphabet 0, 1 that has the set of all strings beginning with 101.

55. Define DFA and NFA. Write any two differences between them.

56. Explain Mealy and Moore machines with examples.
Let $L_1 = \{a^n \mid n \geq 0\}$ and $L_2 = \{b^n \mid n \geq 0\}$.
57. Find and explain:
- $L_1 \cup L_2$
 - $L_1 L_2$ (concatenation)
 - Whether the result is finite or infinite
58. Explain the difference between:
- ϵ (empty string)
 - $\emptyset$ (empty set)
- with examples.
59. Let $\Sigma = \{a, b\}$. List all strings of length ≤ 2 .
Also identify which are valid strings of the language.
60. Describe Kleene Star (*) and Kleene Plus (+) with suitable examples.
61. Given $\Sigma = \{a, b\}$, define:
- Σ^*
 - Σ^+
 - Language of all strings with even length
62. Explain the concept of length of a string. What is the length of ϵ (empty string)? Give examples.
63. Convert Regular Expression $(a+b)^c$ into NFA.
64. Prove that $L = \{a^n b^n \mid n \geq 0\}$ is not a regular language using Pumping Lemma.
65. Prove that $L = \{a^n b^n\}$ is not regular.
66. Define Closure Properties of Regular Languages with example.
67. What is Pumping Lemma for Regular Languages?
68. Explain Arden's Theorem.
69. Define Regular Expression with an example.
70. Construct a DFA and find the number of final and non-final states by using the Language,
 $L = \{b^n \mid n > 0 \text{ \& } n \neq 1, 3, 5\}$

71. Construct a NFA by using the Language $L = \{ \text{Set of all string over } \{0,1\} \text{ containing atleast two } 0's \}$ also find the number of final and non-final states.
72. What is the difference between Mealy and Moore Machine.
73. Construct a DFA by using the Language $L = \{ W \in \{0,1\}^* \mid \# a(W) \bmod 3 \}$
74. What is the difference between DFA and NFA.
75. Define DFA and NFA with example.
76. Differentiate between Terminal and Non-Terminal symbols.
77. Define Regular Grammar with an example.
78. What is a Grammar? Explain its components with an example.
79. Explain Chomsky Hierarchy.
80. Construct a DFA by using the Language $L = \{ a^n \mid n \geq 0 \ \& \ n \neq 3 \}$ and show transition table.
81. Explain string and language. Generate some strings by using this Language $L = \{ W1 \mid W \in \{0,1\}^* \}$
82. Explain reverse of a string and language with suitable examples.
83. Differentiate between recursive and recursively enumerable languages.
84. Design TM for the language $L = \{ a^n b^n c^n \mid n \geq 1 \}$. Explain the concept of decidability.
Prove that the acceptance problem of TM is undecidable.
85. Define decidable and undecidable problems with examples.
86. Define Turing Machine. Explain about model of Turing Machine
87. Convert the following Context Free Grammar to Push Down Automata.
- $$S \rightarrow aAA$$
- $$A \rightarrow aS \mid bS \mid a$$
- Prove that CFLs are not closed under complement using closure properties.
88. Find a CFG, without ϵ productions, unit productions and useless productions equivalent to the grammar defined by
- $$S \rightarrow ABaC$$
- $$A \rightarrow BC$$
- $$B \rightarrow b \mid \epsilon$$
- $$C \rightarrow D \mid \epsilon$$
- $$D \rightarrow d$$
89. Show that the grammar is ambiguous.
- $$S \rightarrow 0A \mid 1B$$

$$A \rightarrow 0AA \mid 1S \mid 1$$

$$B \rightarrow 1BB \mid 0S \mid 0$$

90. What is the language generated by the grammar $G=(V,T,P,S)$ where $P=\{S \rightarrow aSb, S \rightarrow ab\}$?

91. Explain acceptance of PDA by final state and by empty stack.

Design a pushdown automaton which accepts the language $L = \{a^n b^n \mid n \geq 1\}$

92. Convert the following right-linear grammar into an equivalent DFA:

$$S \rightarrow aA \mid b$$

$$A \rightarrow aS \mid bA$$

93. Construct a Finite Automata for Regular Expression $RE = 10+(1+0)0^*$.

Prove that regular languages are closed under homomorphism and inverse homomorphism

94. State and prove Pumping Lemma for regular languages. Use it to test whether a given language is regular or not.

Q1

- (a) Define alphabet and language.
- (b) Give an example of a finite language.

Q2

- (a) Define string and empty string.
- (b) Find length of "aabb".

Q3

- (a) What is grammar?
- (b) List components of grammar.

Q4

- (a) Explain Chomsky hierarchy.
- (b) Classify regular and context-free grammars.

Q5

- (a) Define regular grammar.
- (b) Give one example of a regular grammar.

Q6

- (a) Compare all four types of grammars.
- (b) Arrange them in hierarchy with reason.

Q7

- (a) Define DFA.
- (b) List its components.

Q8

- (a) Define NFA.
- (b) Give one difference from DFA.

Q9

- (a) What is transition function?
- (b) Write its form in DFA.

Q10

- (a) What is ϵ -transition?
- (b) Why is it used in NFA?

Q11

- (a) Explain DFA minimization.
- (b) State its purpose.

Q12

- (a) Explain subset construction.
- (b) Why does DFA sometimes have more states than NFA?

Q11

- (a) Define concatenation in regular expressions.
- (b) Find the concatenation of "a" and "b".

Q12

- (a) What is union in regular expressions?
- (b) Write the language represented by $a + b$.

Q13

- (a) Define closure property.
- (b) Is regular language closed under union?

Q14

- (a) State Arden's Theorem.
- (b) Solve $R = aR + b$.

Q15

- (a) What is a finite automaton?
- (b) Write a regular expression for the language of all strings over $\{a, b\}$ ending with "ab".

Q16

- (a) State pumping lemma.
- (b) Prove $\{a^n b^n\}$ is not regular.

Q17

- (a) Define Context-Free Grammar (CFG).
- (b) Write a CFG for the language $\{a^n b^n \mid n \geq 0\}$.

Q18

- (a) What is a parse tree?
- (b) Draw/describe the parse tree for string "ab" using grammar $S \rightarrow aSb \mid \epsilon$.

Q19

- (a) Define Pushdown Automaton (PDA).
- (b) What is the role of stack in PDA?

Q20

- (a) What is ambiguity in CFG?
- (b) Show that the grammar $S \rightarrow S + S \mid S * S \mid a$ is ambiguous using an example.

Q21

- (a) Define Turing Machine.
- (b) List its components.

Q22

- (a) What is a decidable language?
- (b) Give one example.

Q23

- (a) What is the Halting Problem?
- (b) Why is it undecidable?

Q24

- (a) What is a string over an alphabet?
- (b) Generate all strings of length 2 over the alphabet $\{a, b\}$.
- (c) Generate all strings of length ≤ 2 over the alphabet $\{0, 1\}$.

Q25

- (a) Consider the grammar $S \rightarrow aS \mid bS \mid \epsilon$. What type of grammar is this?
- (b) Generate any four strings from this grammar.
- (c) Describe the language generated by this grammar.

Q26

- (a) Define ambiguity in grammar. (3)
- (b) Determine whether the grammar $S \rightarrow aS \mid Sa \mid a$ is ambiguous.
- (c) Justify your answer with two different derivations for the same string.

Q27

- (a) Construct a DFA over $\{0,1\}$ that accepts all strings ending with "01".
- (b) Draw the state diagram.

Q28

- (a) Construct an NFA over $\{a,b\}$ that accepts strings containing at least one "a".
- (b) Draw the transition diagram.

Q29

- (a) Convert the following NFA to DFA

States: $\{q_0, q_1\}$

Alphabet: $\{0,1\}$

Transitions:

$q_0 \rightarrow 0 \rightarrow q_0, q_1$

$q_0 \rightarrow 1 \rightarrow q_0$

$q_1 \rightarrow 1 \rightarrow q_1$

Final state: q_1

- (b) Draw the equivalent DFA

Q30

- (a) Construct a DFA over $\{0,1\}$ that accepts strings having even number of 0's.
- (b) Convert it into equivalent NFA.
- (c) Minimize the DFA and justify your answer.

Q31

- (a) Define the complement of a language.
- (b) If $L = \{a, ab\}$, find the complement of L over $\Sigma = \{a, b\}^*$ (assume strings of length ≤ 2).
- (c) Represent the complement language using a regular expression.

Q32

- (a) Convert the following regular expression into a finite automaton
 $ab^* + b$
- (b) Describe the language accepted by this expression.
- (c) List any five strings accepted by the automaton.

Q33

- (a) What is a derivation tree in CFG?
- (b) Construct a CFG for the language of all palindromes over $\{a, b\}$.
- (c) Generate the string "aba" using your grammar.

Q34

- (a) Define instantaneous description (ID) in PDA.
- (b) Design a PDA for the language $L = \{w \in \{0,1\}^* \mid \text{number of 0's} = \text{number of 1's}\}$.
- (c) Explain how the stack is used in your design.

Q35

- (a) What is a configuration in a Turing Machine?
- (b) Design a Turing Machine to accept strings containing at least one '1'.
- (c) Explain its working with an example.

Q36

- (a) Define recursively enumerable language.
- (b) Differentiate between recursive and recursively enumerable languages.
- (c) Give one example of each.

Q37

- (a) What is left recursion in CFG?
- (b) Remove left recursion from:
 $S \rightarrow Sa \mid b$

Q38

- (a) What is acceptance by empty stack in PDA?
- (b) Differentiate between acceptance by final state and empty stack.

Q39

- (a) Construct a regular expression for strings over $\{0,1\}$ having at least two consecutive 1's.
- (b) Explain the structure of your expression.
- (c) Generate any five valid strings.

Q40

- (a) Define DPDA and NPDA.
- (b) Compare them.
- (c) Which one is more powerful and why?

Q41

- (a) What is a regular set?
- (b) Construct a regular expression for the language over $\{a, b\}$ where every string starts and ends with the same symbol.